

Smarter Energy Auctions: Neural Network Estimation of Counterfactual Bayesian Nash Equilibria

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Abstract

This paper examines optimal auction design in electricity markets. The Bayesian Nash equilibrium in these markets depends critically on the chosen format, and we focus on comparing uniform-price and discriminatory-price mechanisms. Uniform-price auctions offer simplicity and transparency, as all winning bidders are paid the same market-clearing price. In contrast, discriminatory-price auctions are often viewed as less susceptible to strategic manipulation and collusion, since bidders have no incentive to influence the market-clearing price. The debate remains open, and determining which format minimizes expenditures from the auctioneer’s perspective is therefore non-trivial. To contribute to this discussion, we compare the auctioneer’s expenditure under these two format. We estimate counterfactual discriminatory Bayesian Nash equilibria, using neural networks that take firms’ valuation functions as inputs. Because valuations are unobserved, we approximate them using observed bids through a combination of bootstrapping and kernel methods. From these counterfactual equilibria, we compute the implied expenditures under a discriminatory format and compare them to the realized expenditures under the uniform-price scheme. Our study case is the Colombian Electricity Market, which provides new evidence on how auction design shapes bidding behavior and total market costs.

1 Introduction

Auction theory lies at the heart of the study of markets and economics. Today, it is common to see electricity, government bonds, and even spectrum allocated through auctions. Rather than privatizing these services outright, state agencies organize public auctions to ensure their provision. In Colombia’s electricity market, this approach has been in place since 1994, following inefficiencies under government management. Since then, several auction formats have been tested, with the current system resembling a uniform-price auction. Nevertheless, there remains an ongoing debate about whether alternative designs, particularly discriminatory auctions, could generate lower expenditure’s total costs.

Thus, our research question is: Which auction format (uniform-price or discriminatory-price) yields higher profits in Colombia’s electricity market? The answer is not straightforward, as there are inherent trade-offs that can influence total revenues under both formats. The outcome will depend on the relative strength of these effects. In particular, if incentives to underbid are strong enough, the uniform-price format may perform better.

Solving this question is not straightforward, as it requires computing the counterfactual scenario under a discriminatory auction. The challenge lies in the fact that the equilibrium is difficult to compute, involving a system of thousands of equations that is highly intractable system of partial differential equations. To address this, we employ a novel methodology based on neural networks. Specifically, we use the Neural Pseudogradient Ascent (NPGA) method, which can learn ex-ante equilibrium bidding functions in symmetric Bayesian auction games with continuous types and action spaces. A key difficulty, however, is that the inputs to the neural network are the set of true but possibly unobserved valuations. Since these valuations are unknown, we attempt to recover them from the observed bids.

Our contributions are twofold. First, we apply NPGA to the economic analysis of electricity auctions. While [Bichler et al. \(2021\)](#) approach NPGA from a computer science perspective, emphasizing simulations and computational performance, our work provides the first implementation of this neural network algorithm in an electricity market context. Existing counterfactual studies rely on econometric models or other deep learning methods, but none have employed NPGA for this purpose. Second, we examine the conditions under which valuation functions can be recovered and propose a streamlined estimation strategy inspired by [Balat et al. \(2022\)](#). Whereas their approach relies on a structural model with a closed-form profit function for the electricity sector, our methodology prioritizes simplicity and computational tractability.

The remainder of this document is organized as follows. Section 2 provides background

on auctions and discusses the distinction between uniform- and discriminatory-price formats; readers already familiar with these concepts may skip this section. Section 3 introduces the dynamics of the Colombian electricity market auction. Section 4 outlines the motivation for this study, focusing on the trade-offs between the two auction types and the role of neural networks. Section 5 describes the dataset employed. Section 6 presents the model and the associated optimality conditions. Section 7 details the estimation strategy, including the approximation of the valuation distribution and the implementation of the NPGA algorithm. Finally, Section 8 reports preliminary simulation results.

2 Background

2.1 Auctions

Auctions are mechanisms that allocate goods and determine prices based on participants' bids. They are used in a wide range of applications, including art and housing markets, as well as spectrum and energy allocation. Foundational works include [Vickrey \(1961\)](#), who introduced the second-price auction; [Milgrom and Weber \(1982\)](#), who characterized equilibrium and efficiency under incomplete information; and [Klemperer \(1999\)](#), who linked auction design to market competition.

We focus on environments with incomplete information, modelled as Bayesian games in which bidders' valuations are private. Bidders may participate on either side of the market, buying or selling the product, as described below

2.1.1 Bidders Compete To Buy

The auctioneer offers a good or set of goods, and bidders compete to purchase it. Each bidder has a private valuation and submits a bid. Bids are ranked from highest to lowest, and goods are allocated accordingly until supply is exhausted. The *clearing price* is the lowest accepted bid. Examples include art auctions, real estate sales, and treasury auctions.

2.1.2 Bidders Compete To Sell

The auctioneer seeks to purchase a good supplied by multiple sellers. Each seller submits a bid, typically reflecting marginal cost. Bids are ranked from lowest to highest, and the auctioneer buys sequentially until demand is met. The *clearing price* is the highest accepted bid. This reverse-auction format, also governed by incomplete information, is common in energy procurement, public contracting, and spectrum allocation. In what follows, we focus on this buyer-side auction framework.

2.2 Uniform versus Discriminatory Debate

Throughout the history of auctions, there has been debate about which format should be used. The two main formats are the uniform and the discriminatory designs. In the standard uniform-price auction, also known as pay-as-clear, all bidders are paid the market clearing price, regardless of the bid they originally submitted. In contrast, in the standard discriminatory auction, also known as pay-as-bid, each bidder is paid the price of their own accepted bid.

The central question in this debate is which mechanism provides greater benefits for the auctioneer. This discussion has taken place not only in Colombia, but also worldwide. The choice between these two pricing mechanisms has been a central issue in energy economics and market design. Today, however, the uniform-price format has largely dominated over pay-as-bid in both U.S. and European energy markets, with the main argument being that pay-as-clear promotes truthful bidding ([Caragiannis et al., 2025](#)).

In addition, the European Commission on electricity market design also argue in favour of the uniform-price format:

“This model (pay-as-clear) provides efficiency, transparency and incentives to keep costs as low as possible. [...] The alternative would not provide cheaper prices. In the pay-as-bid model, producers (including cheap renewables) would simply bid at the price they expect the market to clear, not at zero or at their generation costs. Overall, it is better for consumers to have a transparent model that reveals the true costs of energy and provides incentives for individuals to become active in generating their own electricity.”([European Commission, DG Energy, 2025](#))

In general, few European countries currently use strictly discriminatory (pay-as-bid) auctions for electricity. Instead, most follow uniform-pricing mechanisms. Although some markets, such as the UK during certain periods, experimented with or considered pay-as-bid mechanisms, most ultimately converged toward uniform-price systems..

By contrast, in March 2001, the UK adopted a discriminatory or “pay-as-bid” format. British regulators believed that uniform auctions were more susceptible to strategic manipulation by large traders, and expected the new market design to yield substantial reductions in wholesale electricity prices. Similarly, the now-defunct California Power Exchange commissioned a report by leading auction theorists on the advisability of switching to a discriminatory auction for its day-ahead market, following recurrent price spikes in both peak and off-peak periods. The recommendation again was that a discriminatory auction could reduce

the exercise of market power to which the market appeared vulnerable ([Fabra et al., 2002](#)).

Other arguments highlight how discriminatory auctions may reduce price volatility, promote greater competition, or lower average prices ([Rassenti et al., 2001](#)), as well as how they may be less vulnerable to implicit collusion ([Federico and Rahman, 2001](#)).

However, there is no consensus on which format is superior. Both auction types are used in financial and other markets, and there exists a vast economic literature devoted to their analysis. In multi-unit settings, the comparison between the two formats is particularly complex. Neither theory nor empirical evidence provides a definitive case that discriminatory auctions outperform uniform auctions in electricity markets—making this an ongoing and controversial issue.

3 Energy Auctions In Colombia

3.1 Origins

Before the early 1990s, production of energy was handled by state-owned enterprises. Nevertheless, some inefficiencies arose and the system was reformed to an auction system. The political trigger that gave momentum to the reform was the electricity rationing of 1992–1993. Between January 10 and April 10, 1992, energy shortages reached 958 GWh (11% of demand), and between March 2, 1992 and April 10, 1993, 6,138 GWh (16% of demand). This rationing exposed the sector’s deep management problems and highlighted the fiscal burden for the national government in ensuring its sustainability. ([Briceño, 2006](#))

Against this backdrop, and drawing inspiration from the English electricity sector, Colombian legislation underwent a major institutional reform ([República de Colombia, 1994](#), art. de la Ley 143 de 1994), with the aim of providing users with more reliable and efficient electricity services. At that time, technical experts recommended the introduction of an energy exchange model, particularly a uniform multi-unit auction, similar to the one adopted in countries such as England, where the allocation of energy resources was determined through auction mechanisms. In this way, in 1994 Colombia became the first Latin American country to design its electricity market based on the English model of production allocation and price determination through auctions. ([Balat et al., 2022](#))

3.2 Current Basic Structure

The *Mercado de Energía Mayorista* (MEM), established during the liberalization of Colombia's electricity sector, provides the basic structure of today's wholesale market: a centralized platform where interconnected generators compete to supply national demand.

3.2.1 Agents

Generators (Bidders): Firms that produce electricity, such as hydroelectric, thermal, solar, or wind plants. They aim to sell energy competitively and maximize profits.

Market Operator (Auctioneer): XM administers the wholesale electricity market, determines clearing prices, coordinates generation dispatch through the *Centro Nacional de Despacho* (CND), and oversees settlements under CREG's regulation. It operates on behalf of the buyers, aiming to minimize the total expenditure incurred in the market.

Retailers (Buyers): Primarily electricity retailers who purchase energy in the MEM and resell it to final consumers, including households, firms, and industries. While they do not participate directly in the auction process, they are responsible for paying the generators selected by the auctioneer. For consistency with standard auction terminology, we refer to the auctioneer's total expenditure as the buyers' total expenditure, even though, in the specific context of the Colombian electricity market, these payments are made by the buyers themselves.

3.2.2 Daily Dynamic

Daily auctions are conducted under a uniform-price, multi-unit format administered by XM. Generators submit bids that incorporate marginal production costs and strategic considerations, and the auctioneer determines the least-cost dispatch that satisfies hourly demand. The market-clearing (spot) price is the highest accepted bid among dispatched units, and all dispatched generators are paid this uniform price for each kilowatt-hour supplied. Since 2009, this auction framework has supported reliable provision, efficient allocation, and transparent price formation, with XM acting as a neutral operator responsible for regulatory compliance, system security, and market oversight.

4 Motivation

4.1 The Trade-off

We focus on the setting in which the auctioneer demands a product supplied by multiple sellers. In a *uniform-price auction*, bidders are incentivized to submit lower bids: while bidding lower increases their likelihood of being selected (as the auctioneer prefers to buy at a lower price), the final payment they receive is determined by the market-clearing price rather than by their individual bid (see Section 2.2).

In contrast, under a *discriminatory* (or *pay-as-bid*) auction, bidders tend to submit higher bids, since they are paid exactly what they bid and therefore cannot bid below their marginal costs. As illustrated in Figure 1, the auctioneer’s total expenditure equals areas $A + C$ under the uniform-price auction and $A + B$ under the discriminatory auction.

The uniform-price format thus induces more aggressive bidding, pushing bids closer to marginal costs (with incentives to make them as lower as possible as emphasized by [European Commission, DG Energy \(2025\)](#)), because all winning bidders are ultimately paid the same clearing price. The discriminatory format, by contrast, leads to less aggressive bidding, as payments depend directly on each bidder’s offer.

The relative cost-effectiveness of the two formats depends on the relationship between areas B and C : if $C < B$, the uniform-price auction minimizes auctioneer’s total expenditure; if $B < C$, the discriminatory auction is more efficient from the auctioneer’s perspective.

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¹This is a mirror version of the standard case in which the auctioneer sells a product and bidders demand it. In that setting, uniform auctions typically lead to higher bids, as all buyers pay the same market-clearing price—which is weakly lower than their individual bids—whereas in discriminatory auctions, bidders tend to bid more conservatively, since submitting higher bids increases both their likelihood of winning and the price they must pay, which generally leads them to underbid slightly compared to the uniform format

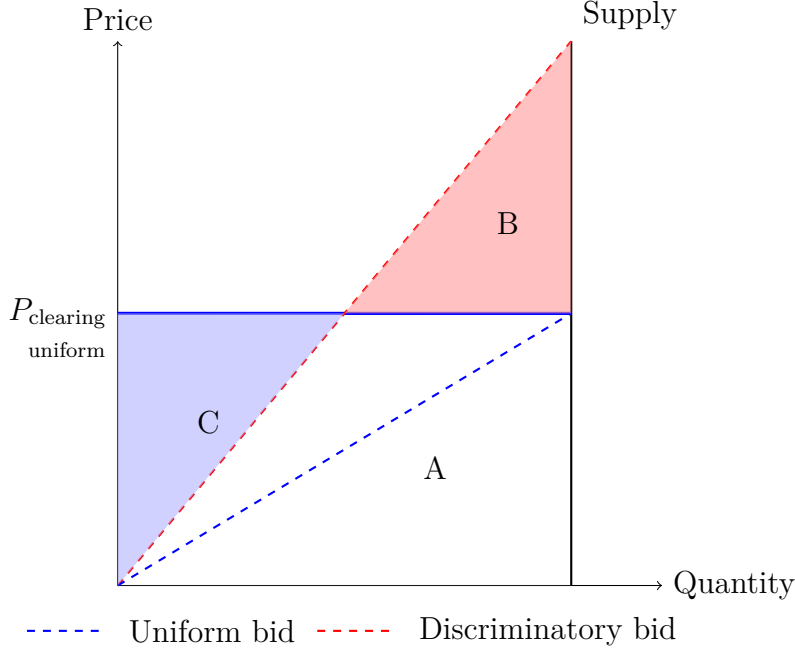


Figure 1: Illustration of the trade-off between uniform and discriminatory auction formats. Bids under the uniform format (dashed blue line) lie below those under the discriminatory format (dashed red line). In the uniform auction, all accepted bidders are paid the same market-clearing price $P_{\text{clearing uniform}}$, implying a total payment of $A + C$ by the auctioneer. In the discriminatory (pay-as-bid) auction, bidders are paid their individual bids, yielding a total payment of $A + B$.

4.2 Neural Networks

The use of neural networks to compute counterfactual outcomes is motivated by the inherent difficulty of solving equilibria in Bayesian auction games analytically. There is limited knowledge regarding Bayesian Nash Equilibrium (BNE) strategies in standard auction formats with multiple objects for sale and interdependent values. Even in single-object auctions, the equilibrium characterization can lead to a system of partial differential equations that generally lacks a closed-form solution. Understanding these environments is crucial, as highlighted by the 2020 Nobel Memorial Prize in Economic Sciences awarded to Paul Milgrom and Robert B. Wilson for their seminal contributions to auction theory with interdependent values ([Janssen, 2020](#)).

Numerical methods for computing BNEs are valuable but often computationally expensive. With the advent of artificial intelligence and machine learning, neural networks have emerged as a promising alternative and have been increasingly used to predict strategic human behavior, forecast auction outcomes in empirical settings, and tackle other problems in

automated market design, such as the discovery of socially optimal tax policies ([Hartford, 2016](#); [Ghani and Simmons, 2004](#); [Zheng et al., 2020](#)).

In this context, neural networks are used to represent the players’ bid functions, with agents learning through self-play. However, standard neural network methods cannot be applied directly because the utility function is discontinuous (see Appendix [A](#)). Consequently, conventional gradient-based approaches (e.g., backpropagation) often fail, frequently resulting in constant-zero bids for all participants. To address this issue, we employ Neural Pseudogradient Ascent (NPGA) as proposed by ([Bichler et al., 2021](#)). The core idea is to compute random perturbations to circumvent the discontinuities, deriving pseudo-gradients via evolutionary strategy optimization rather than relying on exact gradients from standard learning algorithms.

5 Data

For our empirical analysis, we use data on hourly and daily available capacity, daily bid prices, daily market clearing prices, daily aggregate demand, and covariates that characterize the auction, such as fuel prices, water storage, and inflow levels. The sample covers the period from 1 April 2025 to 9 September 2025. All data are provided by XM, except for fuel prices, which are obtained from UPME and are publicly available. The dataset contains 221,832 observations across 79 firms. Figure [2](#) shows the contribution of each generation type, with hydraulic firms dispatching the largest share to meet total demand, followed by thermal generators, and finally solar energy, which is only available during daylight hours. Additional descriptive statistics can be found in Appendix [B](#).

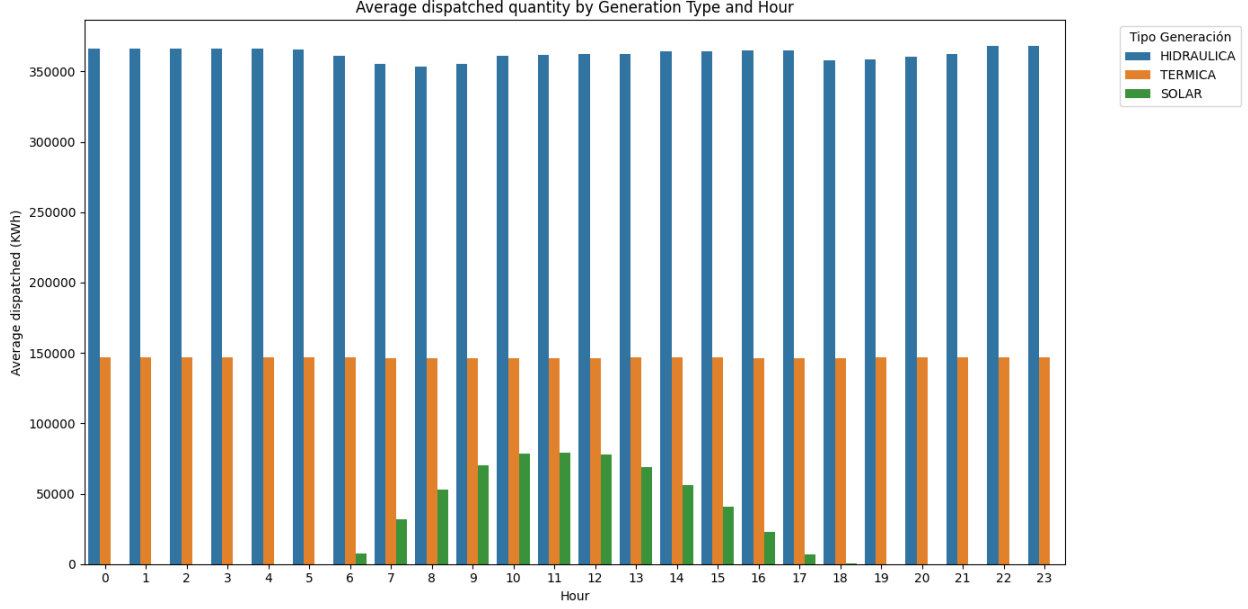


Figure 2: Average contribution in the dispatch per type of generator to total demand

6 Model

In this section, we describe the model that characterizes the behavior of firms in the Colombian Electricity Auction Market, which we are able to estimate using the available data. We derive the optimality conditions for firms' optimal bids (prices) and, subsequently, for their marginal costs.

In our model, there are $i = \{1, \dots, N\}$ firms² each operating a single generation unit during hours $h = \{0, \dots, 23\}$, for each day $t = 1, \dots, T$. Each firm submits a bid vector

$$\vec{b}_i = \{b_i, g_{ih}\}_{h=0}^{23},$$

where b_i is a unique daily price and g_{ih} represents the available (declared) capacity for each hour of the day.

Firms are also assumed to have pre-committed quantities, pos_{ih} , of energy to produce or deliver through bilateral market contracts. To simplify the dynamics of the Colombian Electricity Market, we make the following assumptions:

Assumption 1 (Simple Bids) *Each day, firms submit simple bids consisting of a single daily price and 24 declared capacity values, one for each hour of the day. We assume no start-up costs and no uplift to the hourly price.*

²Also referred to as generators, see 3.2.1

Assumption 2 (Constant Marginal Cost) *The marginal cost is constant and given by:*

$$C_i(Q_i) = \sum_{h=0}^{23} \phi_i Q_{ih}, \quad (1)$$

where ϕ_i denotes the marginal cost of production for firm i .

Assumption 3 (Independent Firms) *Firms operate independently, and each firm is modeled as a single generator.*

Assumption 4 (Rational Bidding) *Firms submit bids rationally to maximize expected profit given market rules and competitors' behavior.*

Assumption 5 (Common Valuation Knowledge) *Firms share common knowledge of the joint distribution of valuations, denoted V , although the realized valuation may differ across firms.*

6.1 Firm's Maximization Problem

The profit function of firm i is given by:

$$\Pi_i(\vec{b}(s)) = \sum_{h=0}^{23} \left[(Q_{ih}(\vec{b}(s)) - \text{pos}_{ih}) p_h(\vec{b}(s)) - C_i(Q_{ih}(\vec{b}(s))) \right], \quad (2)$$

where Q_{ih} denotes the quantity of energy sold in the auction, and pos_{ih} represents the quantity of energy sold under bilateral contracts³ by firm i in hour h ; p_h is the market clearing price under the uniform auction mechanism, and s represents a state or realization from the set S of all feasible dispatches.

In the standard setting, firm i chooses its strategy $\vec{b}_i$ to maximize its expected profit, conditional on the strategies adopted by all other participants, $-i$:

$$\max_{\vec{b}_i} \sum_{s \in S} \Pr(s \mid \vec{b}) \mathbb{E}_{-i} \left[\Pi_i(\vec{b}_i, \vec{b}_{-i}) \mid s \right]. \quad (3)$$

6.2 Optimality Conditions

Note that each firm's optimal choice consists of the vector $\vec{b}_i$, which includes a bid price b_i and a set of quantities $\{g_{ih}\}_{h=0}^{23}$. However, quantities cannot be considered strategic,

³Bilateral contracts are financial contracts between two parties specifying the price and quantity of energy for a defined period. These agreements are executed between generators and *comercializadores*, and are apart from the MEM.

since generators must declare their available capacity in advance according to the schedule established by the *Centro Nacional de Despacho* (CND). Therefore, we focus exclusively on the first-order condition (F.O.C.) with respect to the bid price b_i .

$$\sum_{s \in S} \Pr(s | \vec{b}) \frac{\partial}{\partial b_i} \mathbb{E}_{-i} [\Pi_i(\vec{b}) | s] + \sum_{s \in S} \frac{\partial \Pr(s | \vec{b})}{\partial b_i} \mathbb{E}_{-i} [\Pi_i(\vec{b}) | s] = 0.$$

After some algebra (see Appendix C.1), the optimal bid is given by:

$$b_i^* = \frac{\partial C_i}{\partial Q_{ih}} - \frac{\sum_{h=0}^{23} \mathbb{E}_{-i} [Q_{ih} - \text{pos}_{ih} | s, p_h = b_i]}{\sum_{h=0}^{23} \mathbb{E}_{-i} \left[\frac{\partial Q_{ih}}{\partial b_i} | s, p_h = b_i \right]}. \quad (4)$$

As stated in Assumption 4, each firm chooses its strategy optimally. Consequently, the firm's marginal cost can be directly inferred from its optimal bid. By assumption 2, $\frac{\partial C_i}{\partial Q_{ih}} = \phi_i$ and replacing in equation 4, we obtain:

$$\phi_i = b_i + \frac{\sum_{h=0}^{23} \mathbb{E}_{-i} [Q_{ih} - \text{pos}_{ih} | s, p_h = b_i]}{\sum_{h=0}^{23} \mathbb{E}_{-i} \left[\frac{\partial Q_{ih}}{\partial b_i} | s, p_h = b_i \right]}. \quad (5)$$

As discussed in Section 2.1.2, in energy auctions a firm's marginal cost reflects the minimum price at which it is willing to supply an additional unit of energy. The actual bid price includes a markup over the marginal cost, which depends on the firm's market power on influencing the clearing price: the greater the influence a firm has on the clearing price, the higher the markup it can set.

In essence, $\mathbb{E}_{-i} [Q_{ih} - \text{pos}_{ih} | s, p_h = b_i]$ represents the residual demand curve faced by bidder i and we evaluated at the point when it is the marginal bidder. Meanwhile $\mathbb{E}_{-i} \left[\frac{\partial Q_{ih}}{\partial b_i} | s, p_h = b_i \right]$ is the slope of the of its residual demand curve, which is expected to be negative according to the Demand's Law. Firms with greater market power can increase their bids without substantially reducing their dispatched quantities; that is, bidders with higher markups face a more inelastic residual demand curve near the clearing price. Therefore, $\phi_i \leq b_i$, with equality holding for smaller firms lacking market power.

7 Empirical Strategy

In Section 7.1, we address our first empirical objective which is to approximate the unknown but true distribution of the valuation function from the bidders. This distribution reflects the firms' marginal costs while accounting for uncertainty and incorporating information about their bidding behaviour. Using the observed bid data, we aim to recover the underlying optimization conditions that generated these bids. Under the rationality assumption (Assumption 4), the values of b_i correspond to the *declared prices* submitted by the firms and recorded in the XM database.

In Section 7.2, we address our second empirical objective: to train a neural network capable of characterizing Bayesian Nash Equilibria, thereby facilitating counterfactual analysis. This approach requires as input the firms' valuation functions, which are inherently unobserved. We approximate these valuations using the estimates derived from the observed bid data. Consequently, the neural network receives sample valuations as input and produces optimal bidding strategies consistent with the estimated valuation distributions.

Identifying a true equilibrium necessitates a set of strategies that are mutually consistent, such that each firm's bid is optimal given the bids of all other participants⁴. As discussed in Section 4.2, computing this equilibrium directly is analytically intractable, as it entails solving a system of coupled equations. Therefore, the neural network provides a practical and flexible means to approximate these optimal strategies under strategic interactions.

7.1 Estimation of Firms' Marginal Costs Distribution

To recover empirically the distribution of marginal costs we rewrite equation 5 as:

$$\phi_{it} = b_{it} + \frac{\sum_{h=0}^{23} \mathbb{E}_{-it} \left[Q_{ith} - \text{pos}_{ith} \mid s, p_{ht} = b_{it} \right]}{\sum_{h=0}^{23} \mathbb{E}_{-it} \left[\frac{\partial Q_{ith}}{\partial b_{it}} \mid s, p_{ht} = b_{it} \right]}. \quad (6)$$

We compute $\phi_{i,t}$ (the marginal cost of firm i on day t) rather than a single ϕ_i because the firm faces daily uncertainty in both demand and the strategies of other market participants.

⁴A natural question arises: if the optimal bid formula is already available, why not simply simulate the bids to obtain an equilibrium? The answer is that the formula yields on the marginal costs, which are unknown, and even if we managed to get them, the formula is only a firm's best response conditional on the anticipated strategies of its competitors.

The conditional expectations are computed using a standard bootstrap procedure (see Hortaçsu and Puller (2008), Balat et al. (2022)) summarized as follows:

1. For each fixed day t and each firm i , we take the observed bid of firm i on day t .
2. We identify the set D_{it} of days that are similar to t in terms of relevant covariates for firm i : the price of Fuel Oil, aggregate water stock, river flows, and day-of-the-week indicator. Within this set, we observe the behavior of i 's competitors: their daily prices and quantities.
3. We generate M bootstrap samples from the set D_{it} in which each competitor has an offer (sampling with replacement).
4. For each bootstrap sample $m = 1, \dots, M$, we compute the daily equilibrium (simplified by ordering bids from lowest to highest, following the merit order principle). The market clearing price is denoted by $p_{ht}^{(m)}$.
5. Based on the equilibrium in sample m and i 's rivals' bids, we calculate the conditional dispatched quantity and the derivative using a kernel function ⁵:

$$\hat{Q}_{iht}^{(m)} - \text{pos}_{ih} \Big|_{b_{it}=p_{ht}^{(m)}} \approx D_{ht} - \sum_{k \neq i} g_{kht} K \left(\frac{b_{kth} - p_{ht}^{(m)}}{\gamma} \right),$$

$$\frac{\partial \hat{Q}_{iht}^{(m)}}{\partial b_{it}} \Big|_{b_{it}=p_{ht}^{(m)}} \approx \sum_{k \neq i} g_{kht} \frac{1}{\gamma} K' \left(\frac{b_{kth} - p_{ht}^{(m)}}{\gamma} \right),$$

where $K(u) = \frac{1}{\sqrt{2\pi}} e^{-u^2/2}$, the standard normal density and $K'(u) = -uK(u)$, γ denotes the bandwidth parameter and D_{ht} the total energy demand for hour h during day t .

6. Finally, we average over the M bootstrap samples, obtaining:

$$\hat{\mathbb{E}}_{-it}[Q_{ith} - \text{pos}_{ih} | s, p_{ht} = b_{it}] = \frac{1}{M} \sum_{m=1}^M \hat{Q}_{ith}^{(m)}$$

$$\hat{\mathbb{E}}_{-it} \left[\frac{\partial Q_{ith}}{\partial b_{it}} \Big| s, p_{ht} = b_{it} \right] = \frac{1}{M} \sum_{m=1}^M \frac{\partial \hat{Q}_{ith}^{(m)}}{\partial b_{it}}$$

⁵Since observed values are discrete, small variations in the bid b_{it} can generate arbitrarily large changes in the produced quantity $Q_{ith}^{(m)}$. Even achieving the condition $p_{ht} = b_{it}$ requires M to be sufficiently large, which in turn implies higher computational costs. Using a kernel mitigate these issues and ensure smoothness. (a standard approach; see Appendix D for more detail on the kernel estimation)

After applying this procedure to hour h of each day t and for every firm i , and aggregating by firm and day level, we obtain a time series of marginal cost estimates $\{\phi_{it}\}_{t=1}^T$ according to equation 6. Using Assumption 5, we can recover the underlying valuation function, which we denote by V .

7.2 Neural Network Computation of Counterfactual Bayesian Nash Equilibria

The NPGA method takes the valuation samples v from the true valuation distribution function as input V . Since this is unknown, we approximate it empirically by using the distribution derived from the $\{\phi_{i,t}\}_{t=1}^T$. The NPGA algorithm can be described in summary as follows:

1. Each bidder's bidding function $b_i(v_i)$ is approximated by a neural network $\pi_i(v_i, \theta_i)$, $\theta_i \in \mathbb{R}^{24}$ (it is a multi-unit auction, and our units are the 24- hours of the day)
2. Input layer is sampled valuation $v_i \in V$. Output layer is the respective bid b_i .
3. Objective function is the ex-post utility function of the discriminatory auction.
4. In each simulated auction:
 - (a) Every bidder samples a valuation ϕ_i from V .
 - (b) The network outputs the corresponding bid schedule π_i .
 - (c) Each bidder observes their ex-post utility.
5. After H simulations, players update their parameters θ_i based on the batch of observed utilities.⁶

The detail of the algorithm is presented in E.

8 Results

We characterized the optimality conditions and recovered firms' marginal cost functions. Details on the estimated marginal costs for each firm are presented in Section F. Hydraulic units are generally the most cost-efficient⁷, which is consistent with the historical relevance of hydropower in Colombia.

⁶Assumptions: Bidders are symmetric, risk-neutral, and have independent private values (IPV)

⁷Hydroelectric plants such as Ituango (PES1), Guavio (GVIO), and San Carlos (SNCR) illustrate this pattern.

We then aggregate these results by generation type and report the averages in Figure 3. On average, hydraulic and solar firms exhibit substantially lower marginal costs compared to thermal generation.

When restricting the sample to auction winners, the average marginal cost decreases, as expected in equilibrium, since high-cost firms are less likely to be dispatched. However, because hydro and solar generation alone are not able to meet the total energy demand (as shown in Figure 2) thermal units must also be dispatched.

This pattern extends to markups. Thermal generators not only submit higher bids and operate with higher marginal costs, but they also exhibit some of the largest average markups. This outcome is consistent with the system's reliance on thermal capacity to ensure that demand is fully met. Figure 4 shows that thermal units that clear the market display the highest markups across technologies.

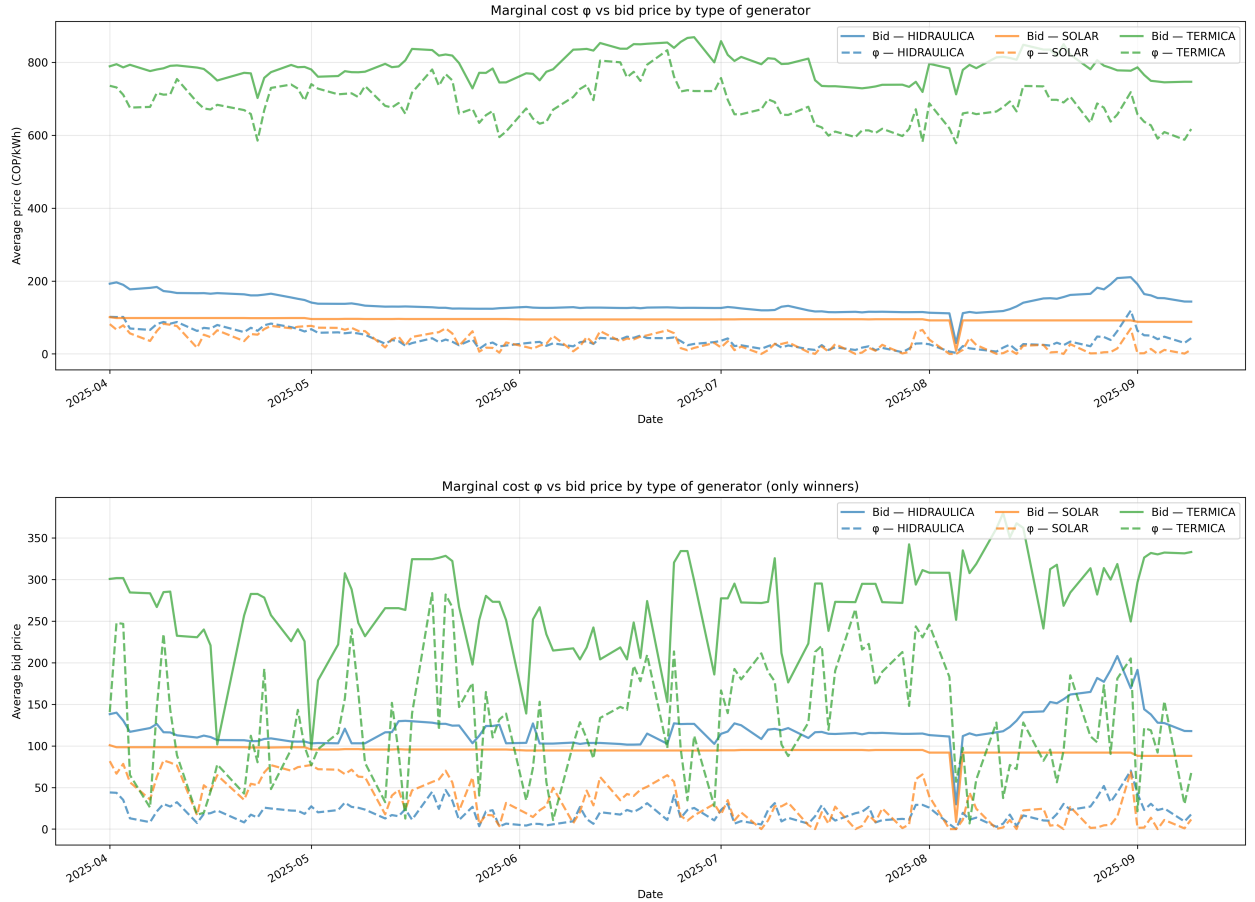


Figure 3: Average bid price and marginal cost for type of generators. Full sample (top) and Winners sample (bottom).



Figure 4: Average markup for type of generators. Full sample (top) and Winners sample (bottom).

9 Preliminary Conclusions

We recover the marginal cost functions from firms' bids in a uniform-price auction. As discussed in earlier sections, a uniform format is preferable to a discriminatory one when the underbidding effect is strong enough; namely, when bids lie close to marginal costs and markups are small enough to compensate the effect of differentiated payments under the discriminatory format. Our results provide two partial but important insights.

First, hydraulic and solar plants are the most cost-efficient technologies but cannot meet total system demand on their own. Consequently, thermal generators must also be dispatched, and these units typically submit both higher bids and have higher marginal costs. In particular, dispatched thermal generators set substantially larger markups than their hydraulic and solar counterparts, as shown in the figures on the previous page.

Second, this naturally shapes the comparison between auction formats. As noted earlier, a central argument for the superiority of uniform pricing is that producers tend to bid at or near their marginal generation costs. However, although Colombian auctions structurally resemble a uniform-price mechanism, thermal units introduce non-trivial markups. This violates the premise underlying the conventional argument, but it does not immediately imply that discriminatory pricing would perform better. On the contrary, this highlights the need to analyze the counterfactual using the neural network in order to quantify the magnitude of the equilibrium effects generated by these markups. The key question is whether the underbidding effect remains strong enough for uniform pricing to outperform discriminatory pricing even in the presence of heterogeneous markups. This motivates the final part of the paper, where we estimate the counterfactual using the neural network.

In summary, if the total demand could be met exclusively by hydraulic and solar generators, their high cost efficiency would strongly favor a uniform-price auction. But this is not the case in Colombia, where thermic generation is necessary. We therefore rely on the counterfactual model to determine whether existing markups are sufficiently small for uniform pricing to remain optimal, or whether discriminatory pricing would dominate due to the discriminatory payment schema.

References

- Balat, J., Carranza, J. E., Martin, J. D., and Riascos, (2022). The effects of changes in the regulation of the colombian wholesale electricity market in a structural model of complex auctions. Technical Report No. 1211, Banco de la República de Colombia.
- Bichler, M., Fichtl, M., Heidekrüger, S., et al. (2021). Learning equilibria in symmetric auction games using artificial neural networks. *Nature Machine Intelligence*, 3:687–695.
- Briceño, A. M. (2006). *Colombia: análisis del sector eléctrico*. Number 4(3) in Serie informes sectoriales. Infraestructura. CAF, Caracas. Retrieved from CAF Scioteca.
- Caragiannis, I., Jiang, Z., and Skoulakis, S. (2025). Department of computer science, aarhus university. Emails: iannis, zhile, stratis@cs.au.dk.
- European Commission, DG Energy (2025). Electricity market design. https://energy.ec.europa.eu/topics/markets-and-consumers/electricity-market-design_en. Accessed: 2025-09-25.
- Fabra, N., von der Fehr, N.-H., and Harbord, D. (2002). Modeling electricity auctions. *The Electricity Journal*, 15(7):72–81. Accessed: 2025-09-25.

- Federico, G. and Rahman, D. (2001). Bidding in an electricity pay-as-bid auction. Working Paper 2001-W5, Nuffield College, Oxford.
- Ghani, R. and Simmons, H. (2004). Predicting the end-price of online auctions. In *Proc. International Workshop on Data Mining and Adaptive Modelling Methods for Economics and Management*. CiteSeer.
- Hartford, J. S. (2016). *Deep Learning for Predicting Human Strategic Behavior*. Ph.d. thesis, University of British Columbia.
- Hortaçsu, A. and Puller, S. L. (2008). Understanding strategic bidding in multi-unit auctions: A case study of the texas electricity spot market. *RAND Journal of Economics*, 39(1):86–114.
- Janssen, M. C. (2020). Reflections on the 2020 nobel memorial prize awarded to paul milgrom and robert wilson. *Erasmus Journal for Philosophy and Economics*, 13:177–184.
- Klemperer, P. (1999). Auction theory: A guide to the literature. *Journal of Economic Surveys*, 13(3):227–286.
- Milgrom, P. R. and Weber, R. J. (1982). A theory of auctions and competitive bidding. *Econometrica*, 50(5):1089–1122.
- Rassenti, S., Smith, V., and Wilson, B. (2001). Discriminatory price auctions in electricity markets: Low volatility at the expense of high price levels. Technical report, Department of Economics, University of Arizona. Working Paper.
- República de Colombia (1994). Ley 143 de 1994: Por la cual se establece el régimen para la generación, interconexión, transmisión, distribución y comercialización de electricidad. <https://www.funcionpublica.gov.co/eva/gestornormativo/norma.php?i=4631>. Expedida el 11 de julio de 1994; Diario Oficial No. 41434.
- Vickrey, W. (1961). Counterspeculation, auctions, and competitive sealed tenders. *Journal of Finance*, 16(1):8–37.
- Zheng, S. et al. (2020). The ai economist: improving equality and productivity with ai-driven tax policies. <https://arxiv.org/abs/2004.13332>.

A Utility function in auction games

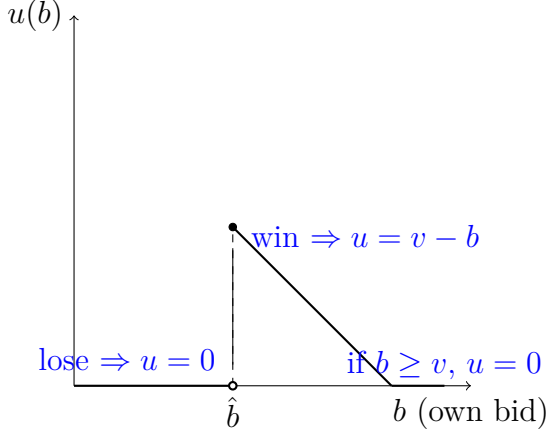


Figure 5: Representation of the discontinuity of utility functions in auction games at $\hat{b}$. v : true valuation, b : bid

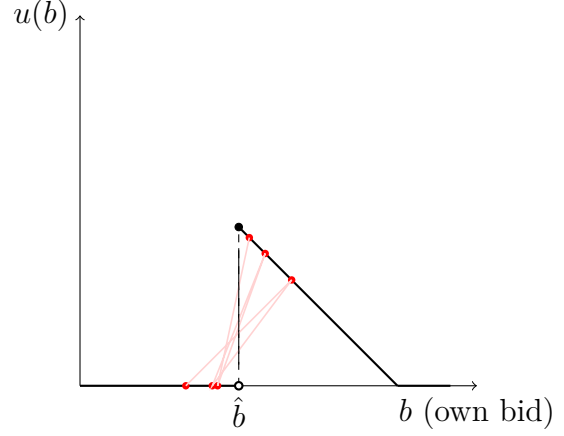


Figure 6: Random jumps (red circles) in NPGA algorithm to avoid getting trapped in the discontinuity.

B Descriptives

Statistic	Bid Price	Available Quantity	Clearing Price	Total Demand
mean	458.018612	3.549415e+08	153.590278	9.465688e+06
std	16.124628	1.566325e+07	127.437081	1.114707e+06
min	379.880659	3.251945e+08	95.060000	6.916520e+06
25%	447.395000	3.427716e+08	105.070000	8.519658e+06
50%	456.108339	3.567482e+08	109.310000	9.510600e+06
75%	467.871793	3.665501e+08	127.380000	1.038452e+07
max	498.835058	3.858039e+08	2224.410000	1.192678e+07

Table 1: Descriptive Statistics 2025-04-01 to 2025-09-09 daily hourly observations. Prices are in COP/kWh and quantities in kWh.

C Derivation of optimality conditions

C.1 Formula for Optimal Bids

The firm's profit function is given by:

$$\Pi_i(\vec{b}(s)) = \sum_{h=0}^{23} \left[(Q_{ih}(\vec{b}(s)) - \text{pos}_{ih}) p_h(\vec{b}(s)) - C_i(Q_{ih}(\vec{b}(s))) \right], \quad (7)$$

where Q_{ih} is the quantity sold by firm i in hour h and p_h is the market clearing price under the uniform auction setting.

Firm i chooses its strategy to maximize its expected profit, conditional on the strategies adopted by all other participants, $-i$:

$$\max_{\vec{b}_i} \sum_{s \in S} \Pr(s \mid \vec{b}) \mathbb{E}_{-i} \left[\Pi_i(\vec{b}_i, \vec{b}_{-i}) \mid s \right]. \quad (8)$$

Note that $\vec{b}_i = \{b_i, g_{ih}\}_{h=0}^{23}$. However, since quantities cannot be strategic due to regulatory constraints, we focus only on the first-order condition (F.O.C.) with respect to the price b_i :

$$\sum_{s \in S} \Pr(s \mid \vec{b}) \frac{\partial}{\partial b_i} E_{-i} \left[\Pi_i(\vec{b}) \mid s \right] + \sum_{s \in S} \frac{\partial \Pr(s \mid \vec{b})}{\partial b_i} E_{-i} \left[\Pi_i(\vec{b}) \mid s \right] = 0$$

Because of the dynamic of Colombian dispatch market as well as Reguant (2014), we assume:

$$\sum_{s \in S} \frac{\partial \Pr(s \mid \vec{b})}{\partial b_i} E_{-i} \left[\Pi_i(\vec{b}) \mid s \right] \approx 0$$

Hence

$$\sum_{s \in S} \Pr(s \mid \vec{b}) \frac{\partial}{\partial b_i} E_{-i} \left[\Pi_i(\vec{b}) \mid s \right] = 0$$

Now:

$$\frac{\partial}{\partial b_i} E_{-i} \left[\Pi_i(\vec{b}) \mid s \right] = E_{-i} \left[\frac{\partial}{\partial b_i} \Pi_i(\vec{b}) \mid s \right].$$

We know that:

$$\Pi_i(\vec{b}(s)) = \sum_{h=0}^{23} \left[\left(Q_{ih}(\vec{b}(s)) - \text{pos}_{ih} \right) p_h(\vec{b}(s)) - C_i(Q_{ih}(\vec{b}(s))) \right]$$

For simplicity, we re-write $b(s) = b$. Then:

$$\begin{aligned} \frac{\partial}{\partial b_i} \Pi_i(b) = & \sum_{h=0}^{23} \left[\frac{\partial}{\partial b_i} \left(Q_{ih}(\vec{b}) - \text{pos}_{ih}(\vec{b}) \right) p_h(\vec{b}) \right. \\ & \left. - \frac{\partial C_i(Q_{ih}(\vec{b}))}{\partial Q_{ih}} \frac{\partial Q_{ih}(\vec{b})}{\partial b_i} \right] \end{aligned}$$

Looking at the first term, we get:

$$\begin{aligned} \frac{\partial}{\partial b_i} \left(Q_{ih}(\vec{b}) - \text{pos}_{ih} \right) p_h(\vec{b}) &= \frac{\partial(Q_{ih}(\vec{b}) - \text{pos}_{ih})}{\partial b_i} p_h(\vec{b}) \\ &+ (Q_{ih}(\vec{b}) - \text{pos}_{ih}) \frac{\partial p_h(\vec{b})}{\partial b_i}. \end{aligned}$$

Hence:

$$\begin{aligned} \frac{\partial}{\partial b_i} E_{-i} \left[\Pi_i(\vec{b}) \mid s \right] &= \sum_{h=0}^{23} E_{-i} \left[\frac{\partial(Q_{ih}(\vec{b}) - \text{pos}_{ih})}{\partial b_i} p_h(\vec{b}) \right. \\ &+ (Q_{ih}(\vec{b}) - \text{pos}_{ih}) \frac{\partial p_h(\vec{b})}{\partial b_i} \\ &\left. - \frac{\partial C_i}{\partial Q_{ih}} \frac{\partial Q_{ih}(\vec{b})}{\partial b_i} \mid s \right] \end{aligned}$$

We know the F.O.C:

$$\sum_{s \in S} \Pr(s \mid \vec{b}) \frac{\partial}{\partial b_i} E_{-i} \left[\Pi_i(\vec{b}) \mid s \right] = 0$$

Then:

$$\sum_{s \in S} \Pr(s \mid \vec{b}) \sum_{h=0}^{23} E_{-i} \left[\frac{\partial(Q_{ih}(\vec{b}) - \text{pos}_{ih})}{\partial b_i} p_h(\vec{b}) + (Q_{ih}(\vec{b}) - \text{pos}_{ih}) \frac{\partial p_h(\vec{b})}{\partial b_i} - \frac{\partial C_i}{\partial Q_{ih}} \frac{\partial Q_{ih}(\vec{b})}{\partial b_i} \mid s \right] = 0$$

Since the condition must hold for every state $s \in S$ and the probability $\Pr(s \mid \vec{b})$ is nonzero, we can simplify the expression by removing the summation over states and the conditional probability.

$$\sum_{h=0}^{23} E_{-i} \left[\frac{\partial(Q_{ih}(\vec{b}) - \text{pos}_{ih})}{\partial b_i} p_h(\vec{b}) + (Q_{ih}(\vec{b}) - \text{pos}_{ih}) \frac{\partial p_h(\vec{b})}{\partial b_i} - \frac{\partial C_i}{\partial Q_{ih}} \frac{\partial Q_{ih}(\vec{b})}{\partial b_i} \mid s \right] = 0$$

At the optimal point, the auction offer $p_h(\vec{b})$ coincides with agent i 's bid, b_i . Therefore, $\frac{\partial p_h(\vec{b})}{\partial b_i} = 1$. At this stage of optimization, agent i is bidding at a price that makes them a “winner” at the margin. Moreover, the derivative of pos_{ih} with respect to b_i is zero, since this quantity corresponds to bilateral contracts previously established. Consequently we consider it exogenous and independent from b_i in our model.

$$\sum_{h=0}^{23} E_{-i} \left[\frac{\partial Q_{ih}}{\partial b_i} b_i + (Q_{ih} - \text{pos}_{ih}) - \frac{\partial C_i}{\partial Q_{ih}} \frac{\partial Q_{ih}}{\partial b_i} \middle| s, p_h = b_i \right] = 0$$

We separate the summation and the expectation into individual terms and then rearrange the equation to isolate b_i .

$$\sum_{h=0}^{23} E_{-i} \left[\frac{\partial Q_{ih}}{\partial b_i} b_i \middle| s, p_h = b_i \right] = \sum_{h=0}^{23} E_{-i} \left[\frac{\partial C_i}{\partial Q_{ih}} \frac{\partial Q_{ih}}{\partial b_i} \middle| s, p_h = b_i \right] - \sum_{h=0}^{23} E_{-i} \left[Q_{ih} - \text{pos}_{ih} \middle| s, p_h = b_i \right]$$

We factor out b_i from the first summation:

$$b_i \sum_{h=0}^{23} E_{-i} \left[\frac{\partial Q_{ih}}{\partial b_i} \middle| s, p_h = b_i \right] = \sum_{h=0}^{23} E_{-i} \left[\frac{\partial C_i}{\partial Q_{ih}} \frac{\partial Q_{ih}}{\partial b_i} \middle| s, p_h = b_i \right] - \sum_{h=0}^{23} E_{-i} \left[Q_{ih} - \text{pos}_{ih} \middle| s, p_h = b_i \right]$$

Finally, solving for b_i we get the optimal bid

$$b_i = \frac{\sum_{h=0}^{23} E_{-i} \left[\frac{\partial C_i}{\partial Q_{ih}} \frac{\partial Q_{ih}}{\partial b_i} \middle| s, p_h = b_i \right] - \sum_{h=0}^{23} E_{-i} \left[Q_{ih} - \text{pos}_{ih} \middle| s, p_h = b_i \right]}{\sum_{h=0}^{23} E_{-i} \left[\frac{\partial Q_{ih}}{\partial b_i} \middle| s, p_h = b_i \right]}$$

Hence:

$$b_i = \frac{\sum_{h=0}^{23} E_{-i} \left[\frac{\partial C_i}{\partial Q_{ih}} \frac{\partial Q_{ih}}{\partial b_i} \middle| s, p_h = b_i \right]}{\sum_{h=0}^{23} E_{-i} \left[\frac{\partial Q_{ih}}{\partial b_i} \middle| s, p_h = b_i \right]} - \frac{\sum_{h=0}^{23} E_{-i} \left[Q_{ih} - \text{pos}_{ih} \middle| s, p_h = b_i \right]}{\sum_{h=0}^{23} E_{-i} \left[\frac{\partial Q_{ih}}{\partial b_i} \middle| s, p_h = b_i \right]}$$

C.2 Formula for Marginal Costs

The prices reported in the database correspond to the actual bids (b_i) submitted by the agents. The first-order condition assumes that each agent behaves rationally and that their bid corresponds to the optimal bid. Consequently, the valuation, defined as the marginal

cost $\left(\frac{\partial C_i}{\partial Q_{ih}}\right)$, is obtained from the optimal bidding equation.

Valuations correspond to marginal costs because a firm's valuation in an auction reflects the minimum price at which it is willing to sell. This minimum value is determined by the cost of producing an additional unit of energy, i.e., the marginal cost.

Thus:

$$b_i \left(\sum_{h=0}^{23} E_{-i} \left[\frac{\partial Q_{ih}}{\partial b_i} \mid s, p_h = b_i \right] \right) + \sum_{h=0}^{23} E_{-i} [Q_{ih} - \text{pos}_{ih} \mid s, p_h = b_i] = \sum_{h=0}^{23} E_{-i} \left[\frac{\partial C_i}{\partial Q_{ih}} \frac{\partial Q_{ih}}{\partial b_i} \mid s, p_h = b_i \right]$$

For simplicity, we assume that the marginal cost $\left(\frac{\partial C_i}{\partial Q_{ih}}\right)$ is constant. By equation (2), this corresponds to ϕ_i . Thus:

$$\sum_{h=0}^{23} E_{-i} \left[\frac{\partial C_i}{\partial Q_{ih}} \frac{\partial Q_{ih}}{\partial b_i} \mid s, p_h = b_i \right] = \phi_i \sum_{h=0}^{23} E_{-i} \left[\frac{\partial Q_{ih}}{\partial b_i} \mid s, p_h = b_i \right]$$

Then, in the original equation:

$$\begin{aligned} b_i \left(\sum_{h=0}^{23} E_{-i} \left[\frac{\partial Q_{ih}}{\partial b_i} \mid s, p_h = b_i \right] \right) + \sum_{h=0}^{23} E_{-i} [Q_{ih} - \text{pos}_{ih} \mid s, p_h = b_i] &= \phi_i \sum_{h=0}^{23} E_{-i} \left[\frac{\partial Q_{ih}}{\partial b_i} \mid s, p_h = b_i \right] \\ \phi_i &= \frac{b_i \left(\sum_{h=0}^{23} E_{-i} \left[\frac{\partial Q_{ih}}{\partial b_i} \mid s, p_h = b_i \right] \right) + \sum_{h=0}^{23} E_{-i} [Q_{ih} - \text{pos}_{ih} \mid s, p_h = b_i]}{\sum_{h=0}^{23} E_{-i} \left[\frac{\partial Q_{ih}}{\partial b_i} \mid s, p_h = b_i \right]} \end{aligned}$$

Therefore, our estimating equation is:

$$\boxed{\phi_i = b_i + \frac{\sum_{h=0}^{23} E_{-i} [Q_{ih} - \text{pos}_{ih} \mid s, p_h = b_i]}{\sum_{h=0}^{23} E_{-i} \left[\frac{\partial Q_{ih}}{\partial b_i} \mid s, p_h = b_i \right]}}$$

D Estimation of Markup via kernel

The hourly markup is given by

$$\frac{\mathbb{E}_{-it} \left[Q_{ith} - \text{pos}_{ith} \mid s, p_{th} = b_{ith} \right]}{\mathbb{E}_{-ith} \left[\frac{\partial Q_{ith}}{\partial b_{ith}} \mid s, p_{th} = b_{ith} \right]}$$

The numerator $\mathbb{E}_{-ith} \left[Q_{ith} - \text{pos}_{ith} \mid s, p_{th} = b_{ith} \right]$ captures the expected dispatch quantity for bidder i on day t and hour h . It corresponds to the residual demand $\mathbf{RD}_{ith}(p_{th})$ faced by firm i evaluated at $p_{th} = b_{ith}$. The expectation is taken over $-i$ because residual demand depends on the bids of i 's competitors.

We recover this object empirically using kernel methods following (Hortaçsu and Puller, 2008). However, it differs from the original paper since in our setting we only observe discrete bid and quantity pairs rather than full supply schedules. Since generators in the Colombian electricity market submit step offers instead of continuous supply curves as in the U.S. market, the residual demand curve cannot be directly observed. To address this limitation, we adopt an extension similar to (Balat et al., 2022), computing equilibria from simulations based on observed bids.

Let $\widehat{\mathbf{S}}_{-i}$ denote the estimated supply schedule of firm i 's rivals, and let $\mathbf{D}_{th}$ denote total demand in day t and hour h . We infer the residual demand $\widehat{\mathbf{RD}}_{ith}(p_{th})$ by solving for the market equilibrium under simulated bid profiles. Formally, for a given hour h , we recover our target as:

$$\begin{aligned} \widehat{\mathbb{E}}_{-it} \left[Q_{ith} - \text{pos}_{ith} \mid s, p_{th} = b_{ith} \right] &= \widehat{\mathbf{RD}}_{ith}(p_{th}) \\ &= \mathbf{D}_{th} - \widehat{\mathbf{S}}_{-i}(p_{th}) \\ &= \mathbf{D}_{th} - \sum_{k \neq i \mid p_{ht}^{(m)} \geq b_{kht}} (g_{kht} - \text{pos}_{kth}) K \left(\frac{b_{kht} - b_{iht}}{\gamma} \right) \end{aligned}$$

The last step follows from the kernel approximation. Although many kernel functions could be used, we employ the standard normal kernel, $K(u) = \frac{1}{\sqrt{2\pi}} e^{-u^2/2}$ whose derivative is $K'(u) = -u K(u)$.

To ensure that the dispatched quantity never exceeds the maximum amount offered by

firm i , we apply a minimum operator:

$$\widehat{\mathbb{E}}_{-it} \left[Q_{ith} - \text{pos}_{ith} \mid s, p_{th} = b_{ith} \right] = \min \left\{ g_{ith} - \text{pos}_{kth}, \quad \mathbf{D} - \sum_{k \neq i | p_{ht}^{(m)} \geq b_{kht}} (g_{kht} - \text{pos}_{kth}) K \left(\frac{b_{kht} - b_{iht}}{\gamma} \right) \right\}.$$

The denominator $\mathbb{E}_{-ith} \left[\frac{\partial Q_{ith}}{\partial b_{ith}} \mid s, p_{th} = b_{ith} \right]$ captures the slope of the residual demand curve. Note that this slope is expected to be negative under the Demand's Law. Equivalently, it can be interpreted as the negative slope of the aggregate supply from rival firms. Formally:

$$\begin{aligned} \widehat{\mathbb{E}}_{-it} \left[\frac{\partial Q_{it}}{\partial b_{it}} \mid s, p_t = b_{it} \right] &= \widehat{\mathbf{RD}}'_i(p) \\ &= -\mathbf{S}'_{-i}(p) \\ &= - \left(\sum_{k \neq i} g_{kht} K \left(\frac{b_{kht} - p_{ht}^{(m)}}{\gamma} \right) \right)' \\ &= \sum_{k \neq i} g_{kht} \left[\frac{1}{\gamma} K' \left(\frac{b_{kht} - p_{ht}^{(m)}}{\gamma} \right) \right] \quad \left(\text{Assuming } \frac{\partial p_{ht}^{(m)}}{\partial b_{it}} = 1 \right) \\ &= - \sum_{k \neq i} g_{kht} \left(\frac{b_{kht} - p_{ht}^{(m)}}{\gamma} \right) \left[\frac{1}{\gamma} K \left(\frac{b_{kht} - p_{ht}^{(m)}}{\gamma} \right) \right] \quad \left(\text{Since } K'(u) = -uK(u) \right) \end{aligned}$$

Now, note that there are three cases:

Case 1. $b_{kht} - p_{ht}^{(m)} > 0$: this implies $b_{kht} > p_{ht}^{(m)}$, so generator k did not win and is out of the market. In this case, we are measuring how much firm i can increase its bid before being displaced by k .

Case 2. $b_{kht} - p_{ht}^{(m)} = 0$: this implies generator k is also marginal, for practice its contribution does not matter since its as marginal as i .

Case 3. $b_{kht} - p_{ht}^{(m)} < 0$: generator k won the bid. This does not represent a threat for the marginal bidder i .

We need to note the kernel by itself does not have an economic interpretation, to make it consistent we economy theory and guarantee the demand law, we re-rewrite the equation as:

$$- \sum_{k \neq i | b_{kht} \geq p_{ht}^{(m)}} g_{kht} \left(\frac{b_{kht} - p_{ht}^{(m)}}{\gamma} \right) \left[\frac{1}{\gamma} K \left(\frac{b_{kht} - p_{ht}^{(m)}}{\gamma} \right) \right]$$

E Neural Pseudogradient Ascend Algorithm

Require: Agents $i \in \mathcal{I}$ with initial policies $\beta_i^0 := \pi_i(\cdot; \theta_i^0)$ induced by initial parameters θ_i^0 ;

ES population size P , ES noise standard deviation σ , learning rate η , batch size H

1: **for** $t = 1, 2, \dots$ **do**

2: Sample a batch $(v_h, o_h)_{h=1}^H$ of valuation and observation profiles from the prior f

3: Calculate joint utility in current strategy profile:

$$\tilde{u}^{t-1} := \frac{1}{H} \sum_h \tilde{u}(v_h, \beta^{t-1}(o_h))$$

4: **for** each agent $i \in \mathcal{I}$ **do**

5: Sample P perturbations of agent i 's current policy:

$$\pi_{ip} := \pi_i(\cdot; \theta_p), \quad \theta_p := \theta_i^{t-1} + \epsilon_p, \quad \epsilon_p \sim \mathcal{N}(0, \sigma^2 I)$$

6: **for** each $p \in \{1, \dots, P\}$ **do**

7: Evaluate the fitness of θ_p :

$$\varphi_p := \frac{1}{H} \sum_h u_i(v_h, \pi_{ip}(o_{h,i}), \beta_{-i}^{t-1}(o_{h,-i})) - \tilde{u}_i^{t-1}$$

8: **end for**

9: Calculate ES pseudogradient:

$$\nabla^{ES} \tilde{u}_i^{t-1} := \frac{1}{\sigma^2 P} \sum_p \varphi_p \epsilon_p$$

10: Gradient update:

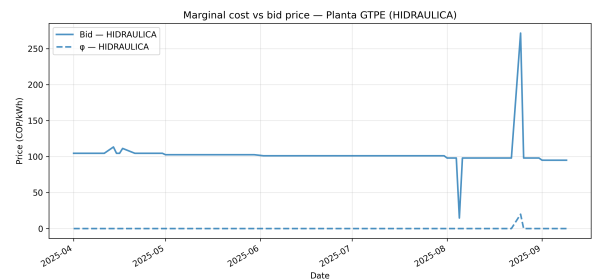
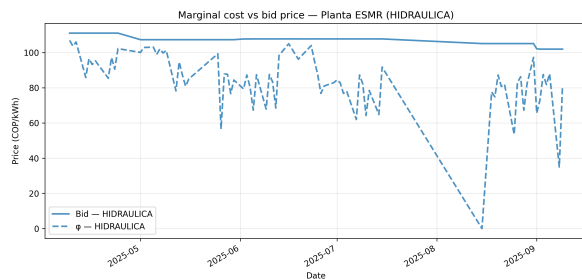
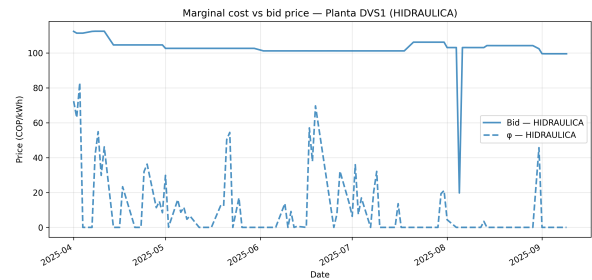
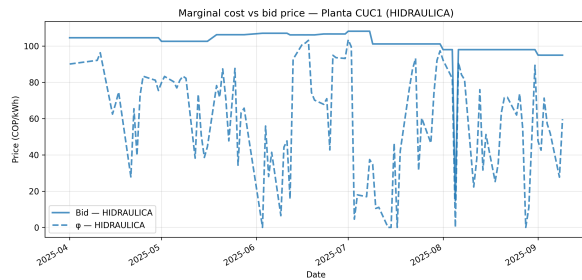
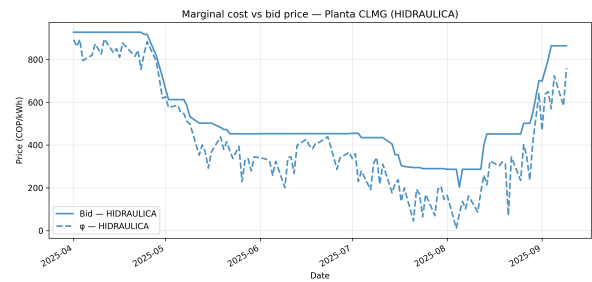
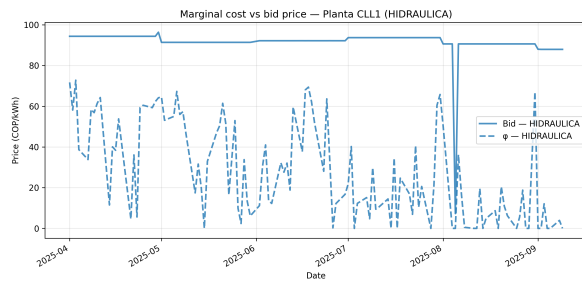
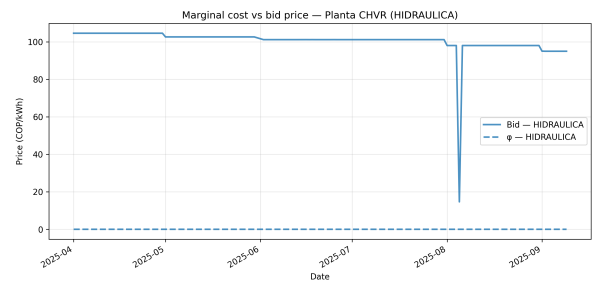
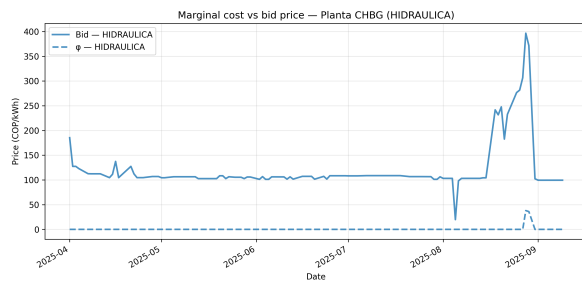
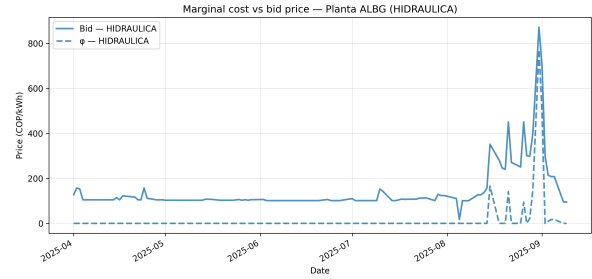
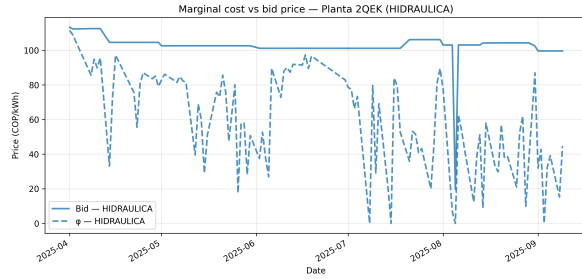
$$\Delta \theta_i^t := \eta \nabla^{ES} \tilde{u}_i^{t-1}, \quad \theta_i^t := \theta_i^{t-1} + \Delta \theta_i^t, \quad \beta_i^t := \pi_i(\cdot; \theta_i^t)$$

11: **end for**

12: **end for**

F Estimated Marginal Cost Per Firm

Hidraulic generation firms



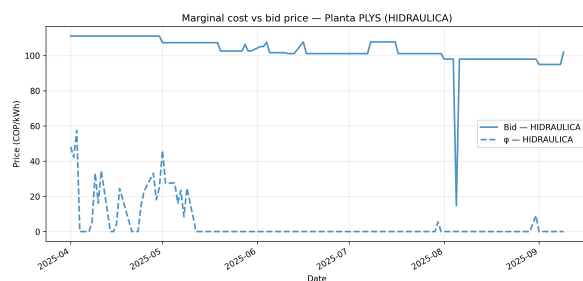
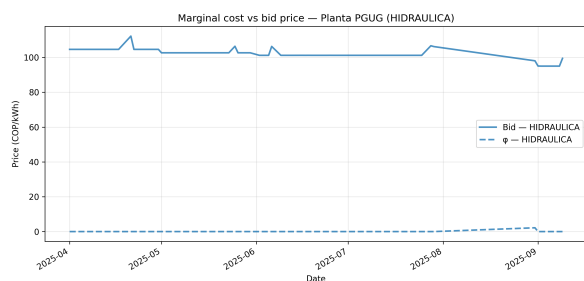
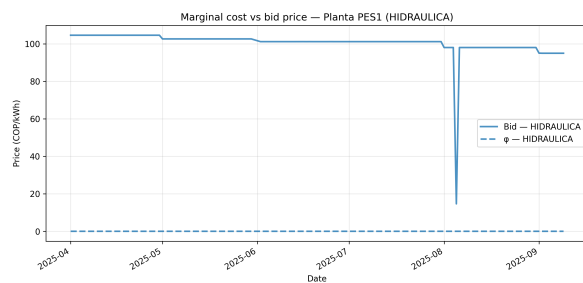
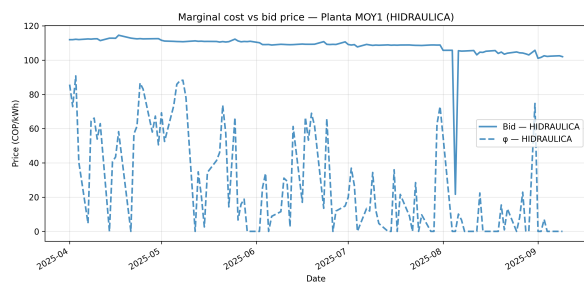
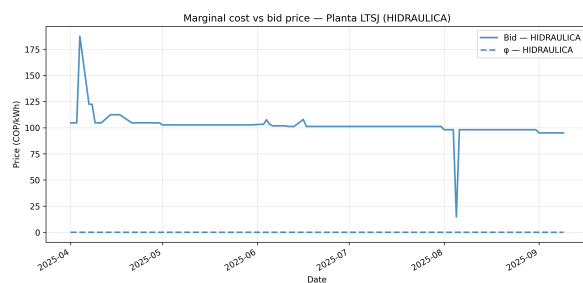
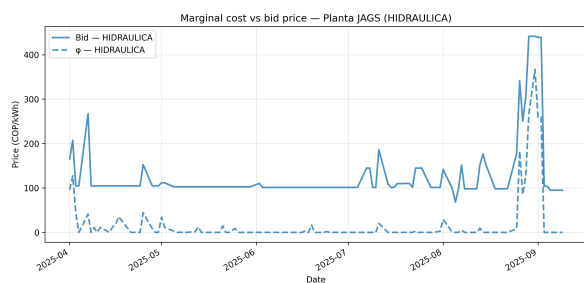
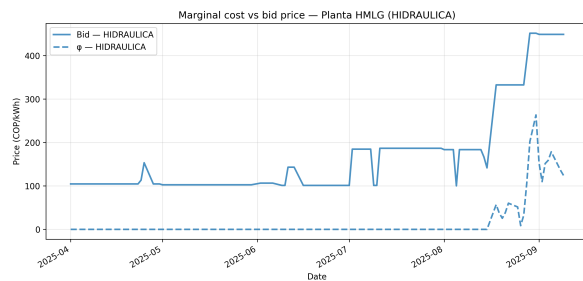
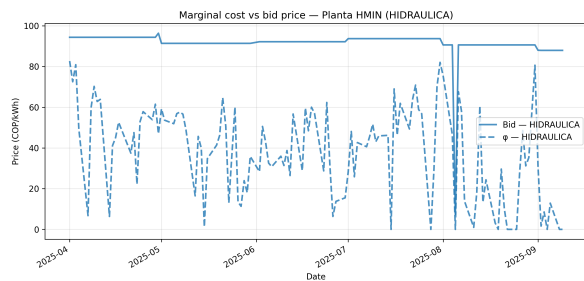
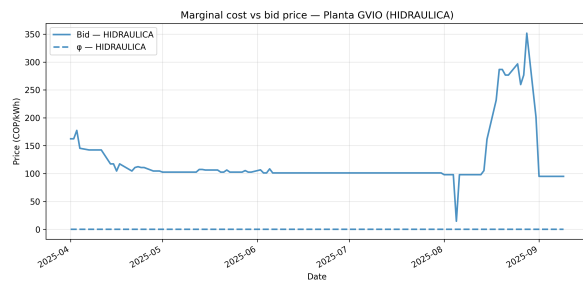
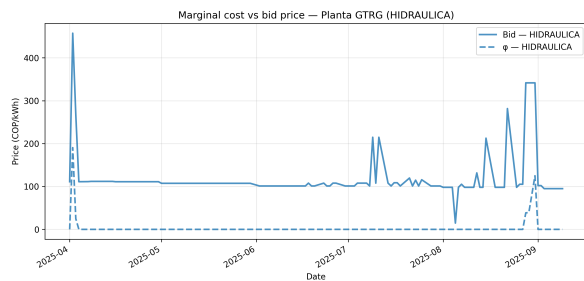




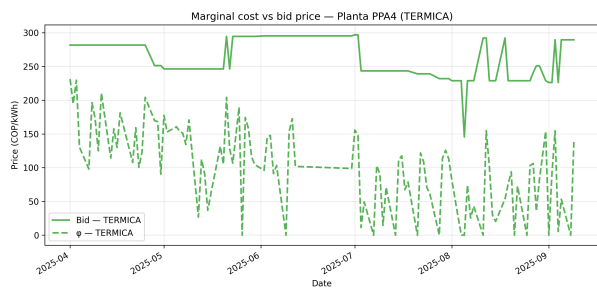
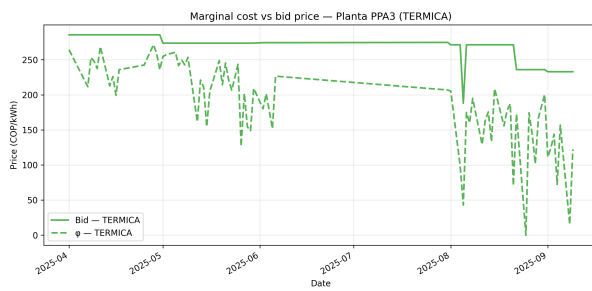
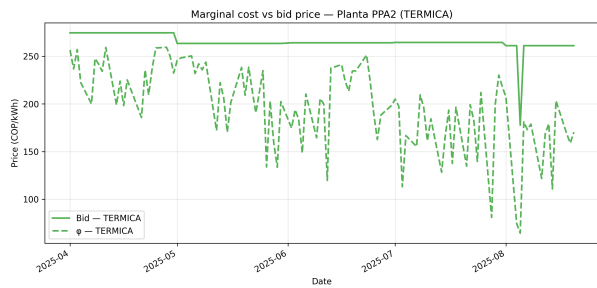
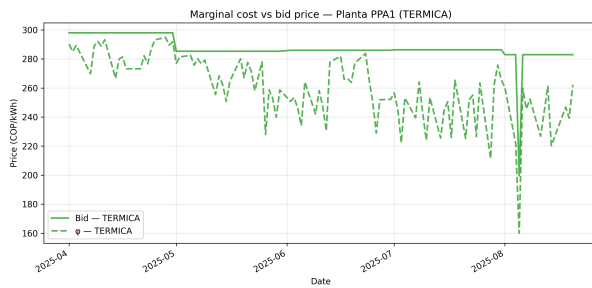
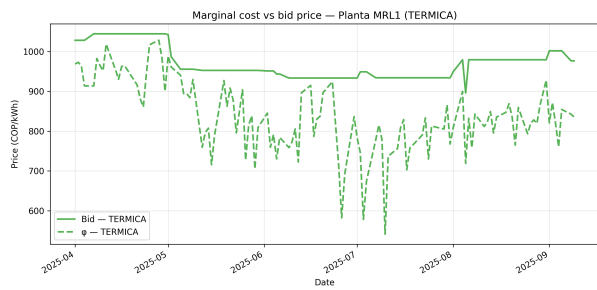
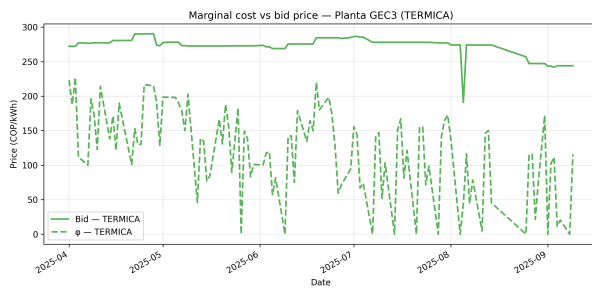
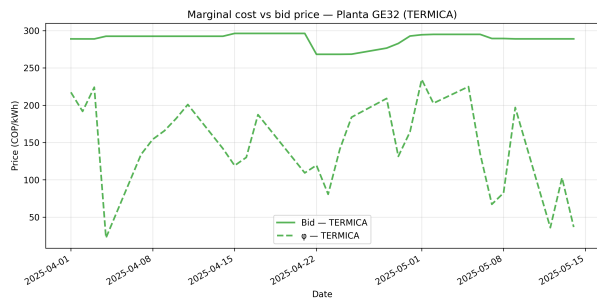
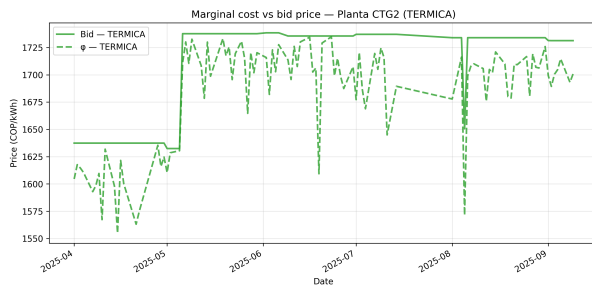
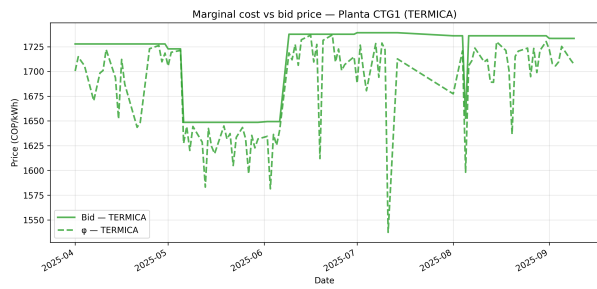
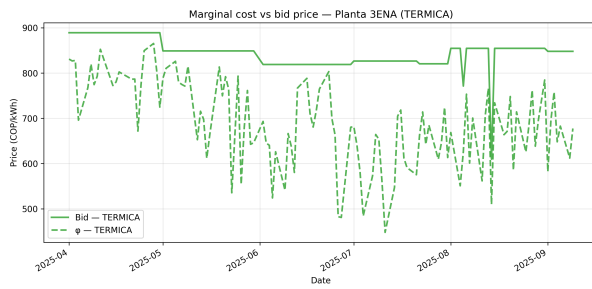
Figure 7: Marginal cost functions hydraulic firms

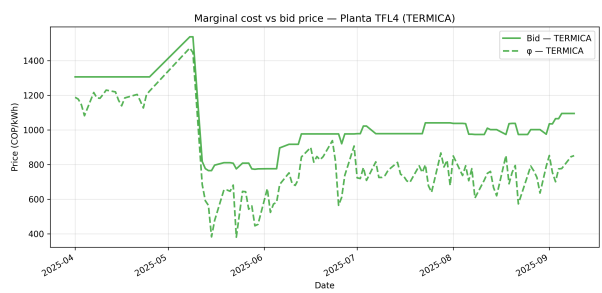
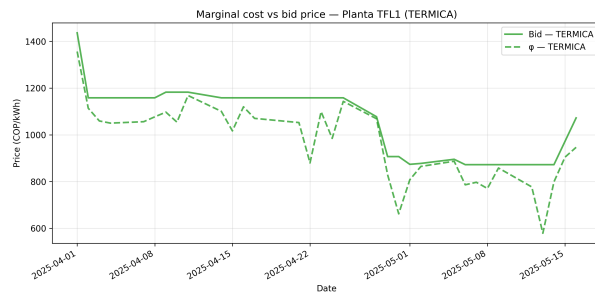
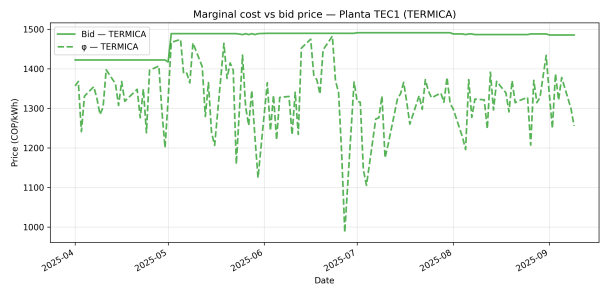
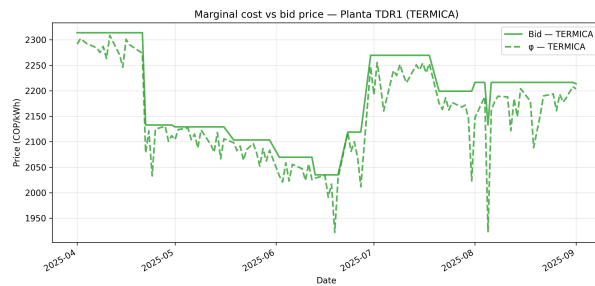
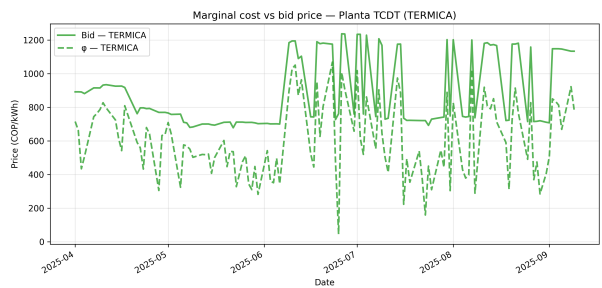
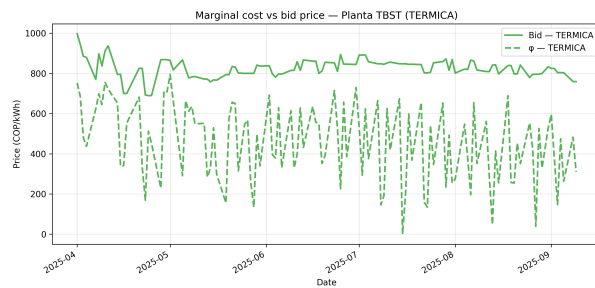
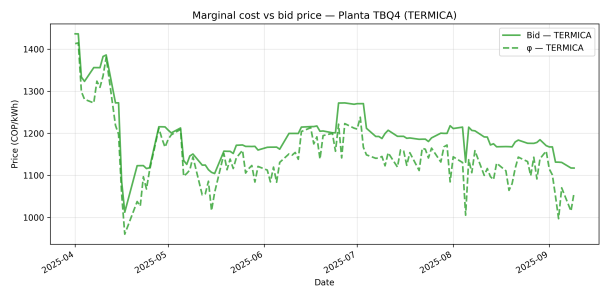
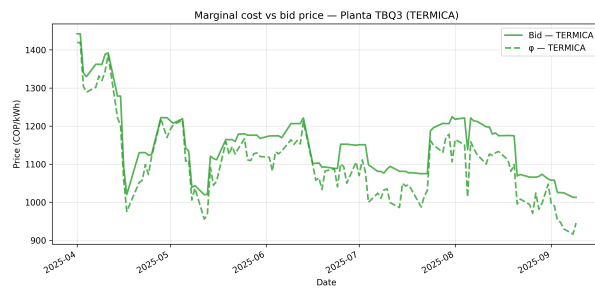
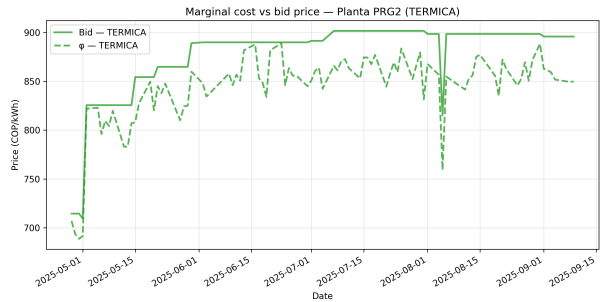
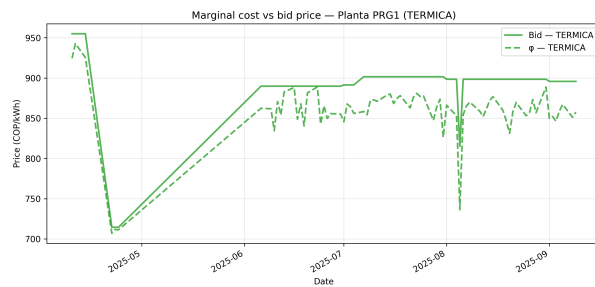
Solar generation firms

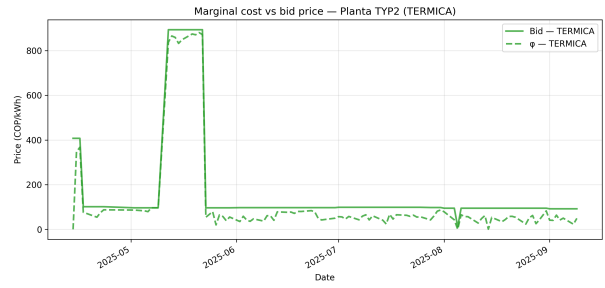
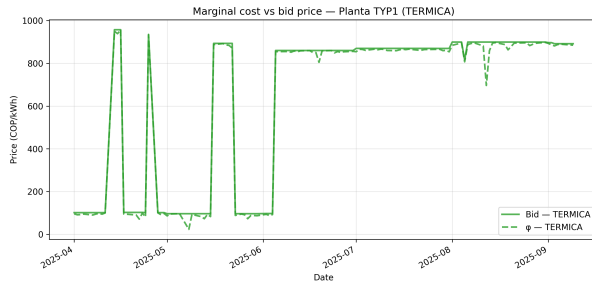
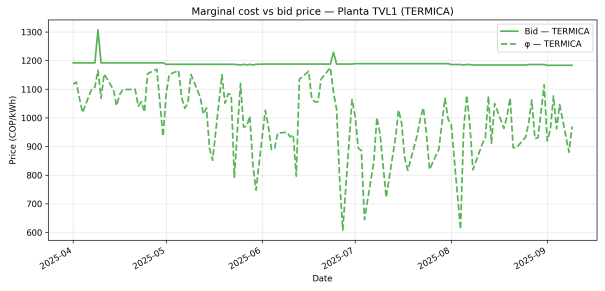
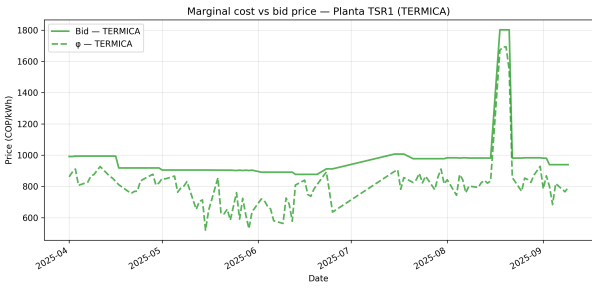
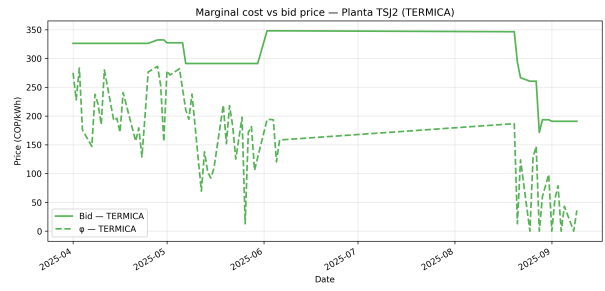
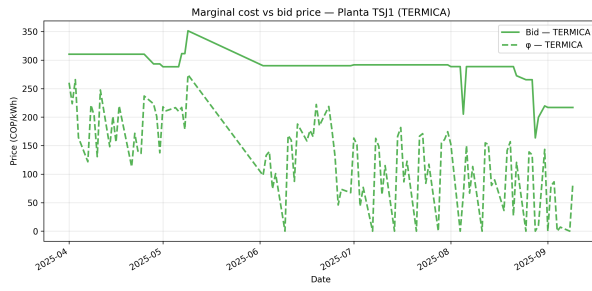
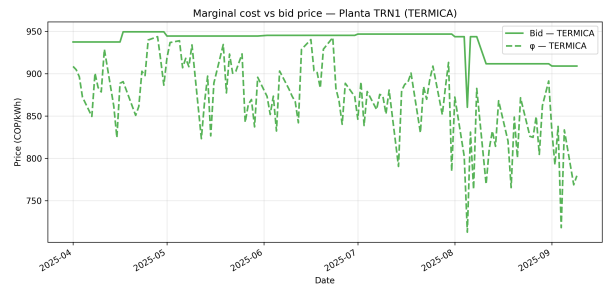
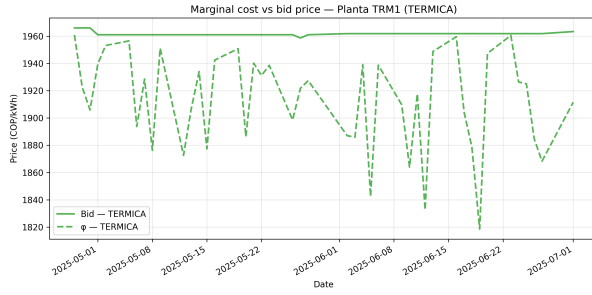
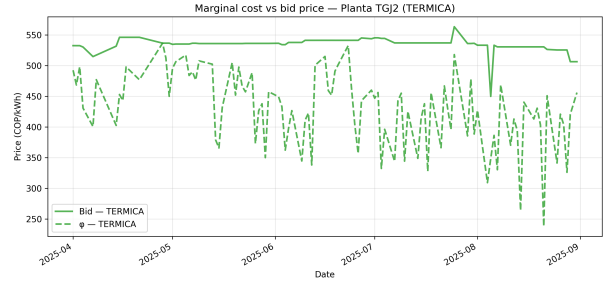
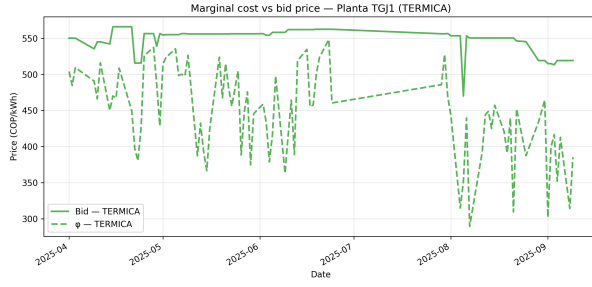


Figure 8: Marginal Cost Solar firms

Thermic generation firms







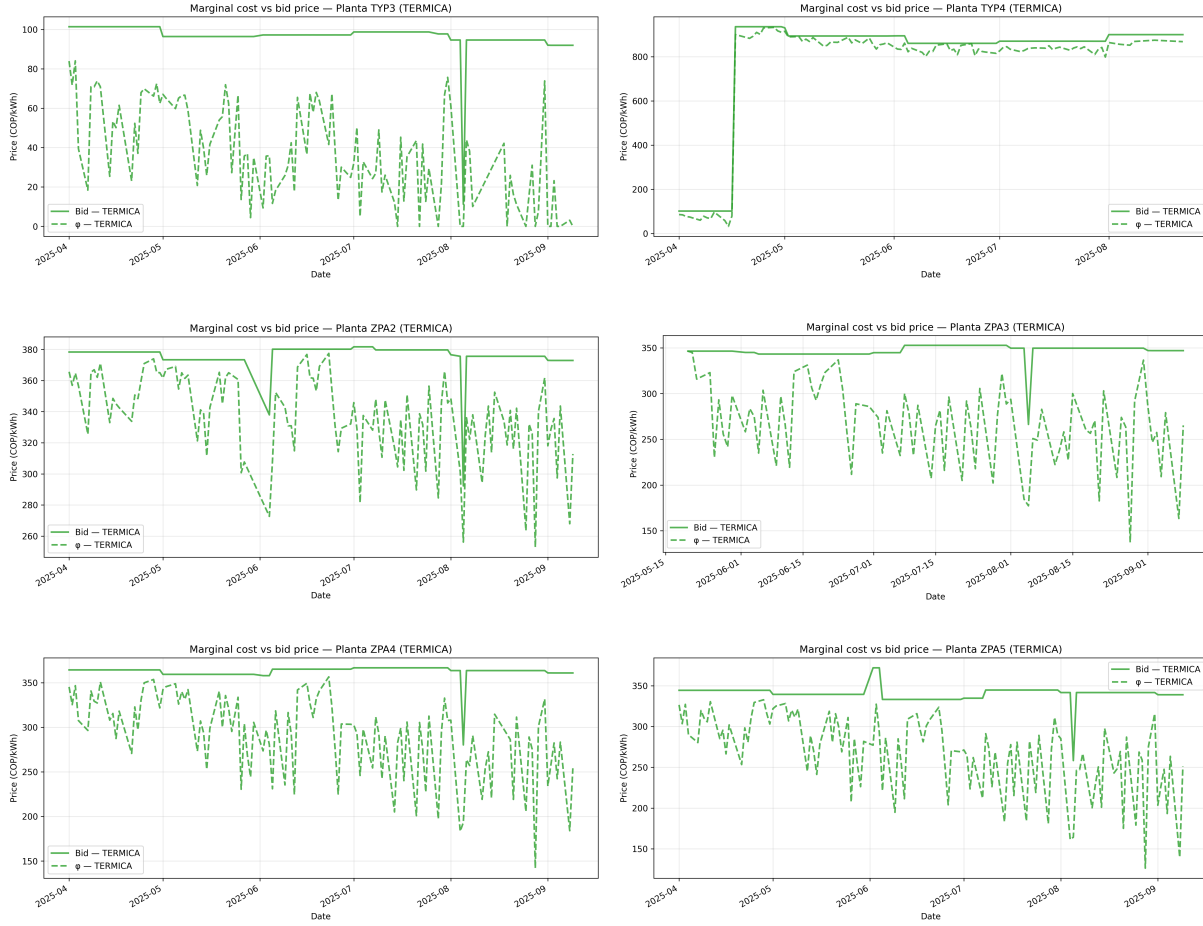


Figure 9: Marginal Cost thermic firms

G Replication Materials

All replication code and instructions are available at <https://github.com/anamariapatron/TesisPEG>